

5 Power controller (PWR)

This section applies to the whole STM32F4xx family, unless otherwise specified.

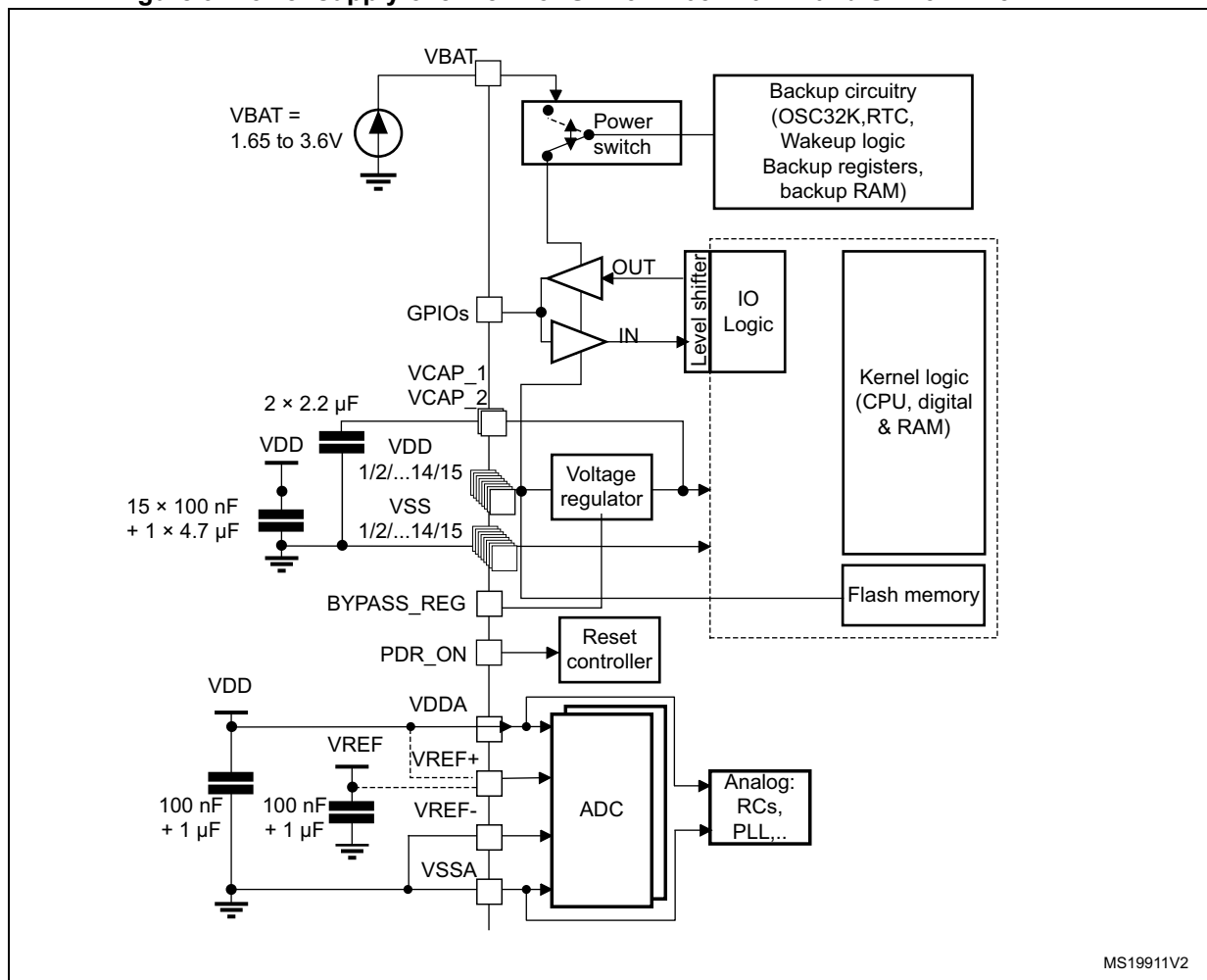
5.1 Power supplies

The device requires a 1.8 to 3.6 V operating voltage supply (V_{DD}). An embedded linear voltage regulator is used to supply the internal 1.2 V digital power.

The real-time clock (RTC), the RTC backup registers, and the backup SRAM (BKP SRAM) can be powered from the V_{BAT} voltage when the main V_{DD} supply is powered off.

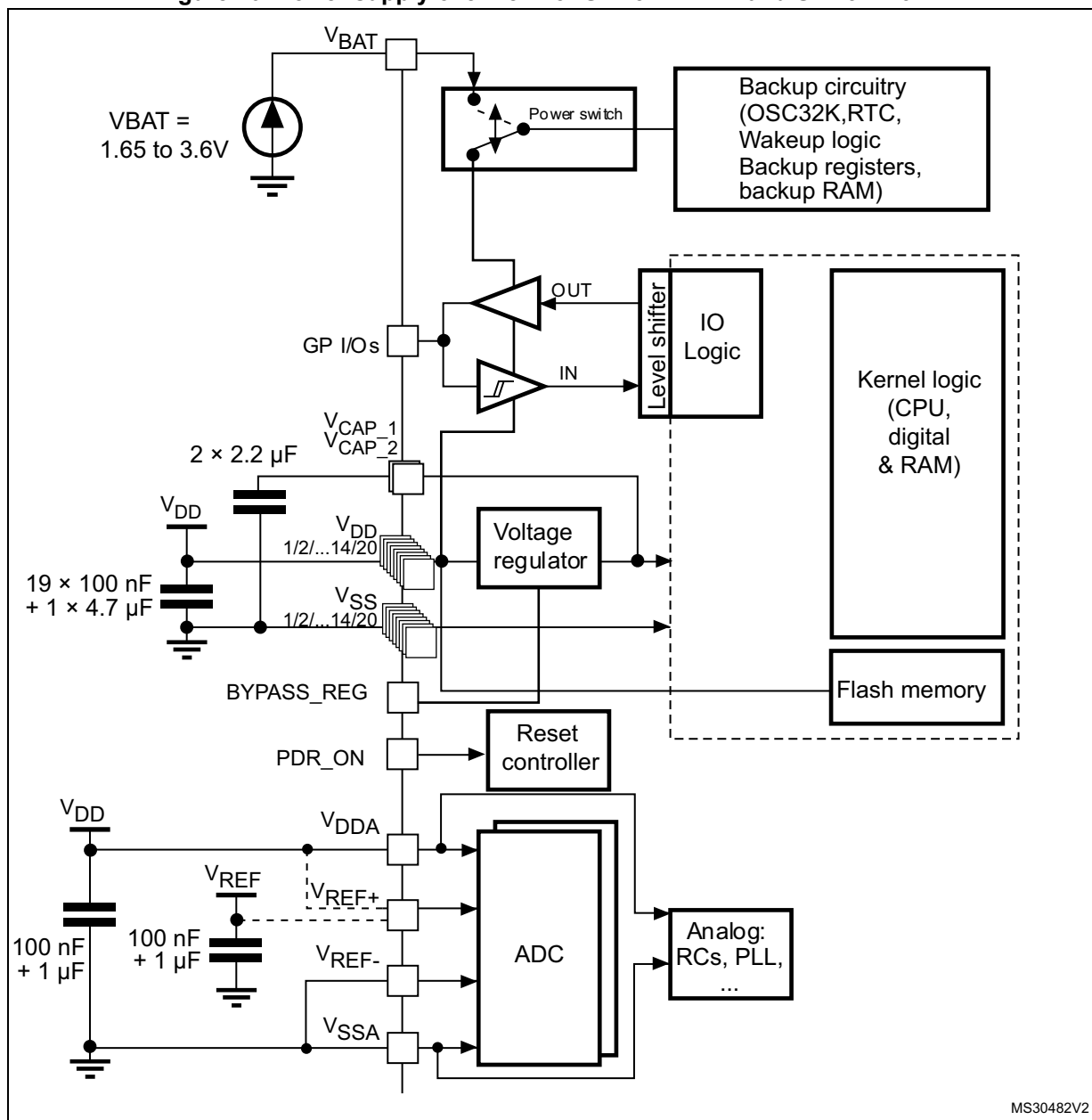
Note: Depending on the operating power supply range, some peripheral may be used with limited functionality and performance. For more details refer to section “General operating conditions” in STM32F4xx datasheets.

Figure 9. Power supply overview for STM32F405xx/07xx and STM32F415xx/17xx



1. V_{DDA} and V_{SSA} must be connected to V_{DD} and V_{SS} , respectively.

Figure 10. Power supply overview for STM32F42xxx and STM32F43xxx



MS30482V2

1. V_{DDA} and V_{SSA} must be connected to V_{DD} and V_{SS} , respectively.

5.1.1 Independent A/D converter supply and reference voltage

To improve conversion accuracy, the ADC has an independent power supply which can be separately filtered and shielded from noise on the PCB.

- The ADC voltage supply input is available on a separate V_{DDA} pin.
- An isolated supply ground connection is provided on V_{SSA} pin.

To ensure a better accuracy of low voltage inputs, the user can connect a separate external reference voltage ADC input on V_{REF} . The voltage on V_{REF} ranges from 1.8 V to V_{DDA} .

5.1.2 Battery backup domain

Backup domain description

To retain the content of the RTC backup registers, backup SRAM, and supply the RTC when V_{DD} is turned off, VBAT pin can be connected to an optional standby voltage supplied by a battery or by another source.

To allow the RTC to operate even when the main digital supply (V_{DD}) is turned off, the VBAT pin powers the following blocks:

- The RTC
- The LSE oscillator
- The backup SRAM when the low-power backup regulator is enabled
- PC13 to PC15 I/Os, plus PI8 I/O (when available)

The switch to the V_{BAT} supply is controlled by the power-down reset embedded in the Reset block.

Warning: During $t_{RSTTEMPO}$ (temporization at V_{DD} startup) or after a PDR is detected, the power switch between V_{BAT} and V_{DD} remains connected to V_{BAT} .

During the startup phase, if V_{DD} is established in less than $t_{RSTTEMPO}$ (Refer to the datasheet for the value of $t_{RSTTEMPO}$) and $V_{DD} > V_{BAT} + 0.6\text{ V}$, a current may be injected into V_{BAT} through an internal diode connected between V_{DD} and the power switch (V_{BAT}).

If the power supply/battery connected to the VBAT pin cannot support this current injection, it is strongly recommended to connect an external low-drop diode between this power supply and the VBAT pin.

If no external battery is used in the application, it is recommended to connect VBAT pin to V_{DD} supply, and add a 100 nF ceramic decoupling capacitor on VBAT pin.

When the backup domain is supplied by V_{DD} (analog switch connected to V_{DD}), the following functions are available:

- PC14 and PC15 can be used as either GPIO or LSE pins
- PC13 can be used as a GPIO as the RTC_AF1 pin (refer to [Table 38: RTC_AF1 pin](#) for more details about this pin configuration)

Note: Due to the fact that the switch only sinks a limited amount of current (3 mA), the use of PI8 and PC13 to PC15 GPIOs in output mode is restricted: the speed has to be limited to 2 MHz with a maximum load of 30 pF and these I/Os must not be used as a current source (e.g. to drive an LED).

When the backup domain is supplied by V_{BAT} (analog switch connected to V_{BAT} because V_{DD} is not present), the following functions are available:

- PC14 and PC15 can be used as LSE pins only
- PC13 can be used as the RTC_AF1 pin (refer to [Table 38: RTC_AF1 pin](#) for more details about this pin configuration)
- PI8 can be used as RTC_AF2

Backup domain access

After reset, the backup domain (RTC registers, RTC backup register and backup SRAM) is protected against possible unwanted write accesses. To enable access to the backup domain, proceed as follows:

- Access to the RTC and RTC backup registers
 1. Enable the power interface clock by setting the PWREN bits in the RCC_APB1ENR register (see [Section 7.3.13](#) and [Section 6.3.13](#))
 2. Set the DBP bit in the [Section 5.4.1](#) and *PWR power control register (PWR_CR) for STM32F42xxx and STM32F43xxx* to enable access to the backup domain
 3. Select the RTC clock source: see [Section 7.2.8: RTC/AWU clock](#)
 4. Enable the RTC clock by programming the RTCEN [15] bit in the [Section 7.3.20: RCC Backup domain control register \(RCC_BDCR\)](#)
- Access to the backup SRAM
 1. Enable the power interface clock by setting the PWREN bits in the RCC_APB1ENR register (see [Section 7.3.13](#) and [Section 6.3.13](#) for STM32F405xx/07xx and STM32F415xx/17xx and STM32F42xxx and STM32F43xxx, respectively)
 2. Set the DBP bit in the *PWR power control register (PWR_CR) for STM32F405xx/07xx and STM32F415xx/17xx* and *PWR power control register (PWR_CR) for STM32F42xxx and STM32F43xxx* to enable access to the backup domain
 3. Enable the backup SRAM clock by setting BKPSRAMEN bit in the [RCC_AHB1 peripheral clock enable register \(RCC_AHB1ENR\)](#).

RTC and RTC backup registers

The real-time clock (RTC) is an independent BCD timer/counter. The RTC provides a time-of-day clock/calendar, two programmable alarm interrupts, and a periodic programmable wake-up flag with interrupt capability. The RTC contains 20 backup data registers (80 bytes) which are reset when a tamper detection event occurs. For more details refer to [Section 26: Real-time clock \(RTC\)](#).

Backup SRAM

The backup domain includes 4 Kbytes of backup SRAM addressed in 32-bit, 16-bit or 8-bit mode. Its content is retained even in Standby or V_{BAT} mode when the low-power backup regulator is enabled. It can be considered as an internal EEPROM when V_{BAT} is always present.

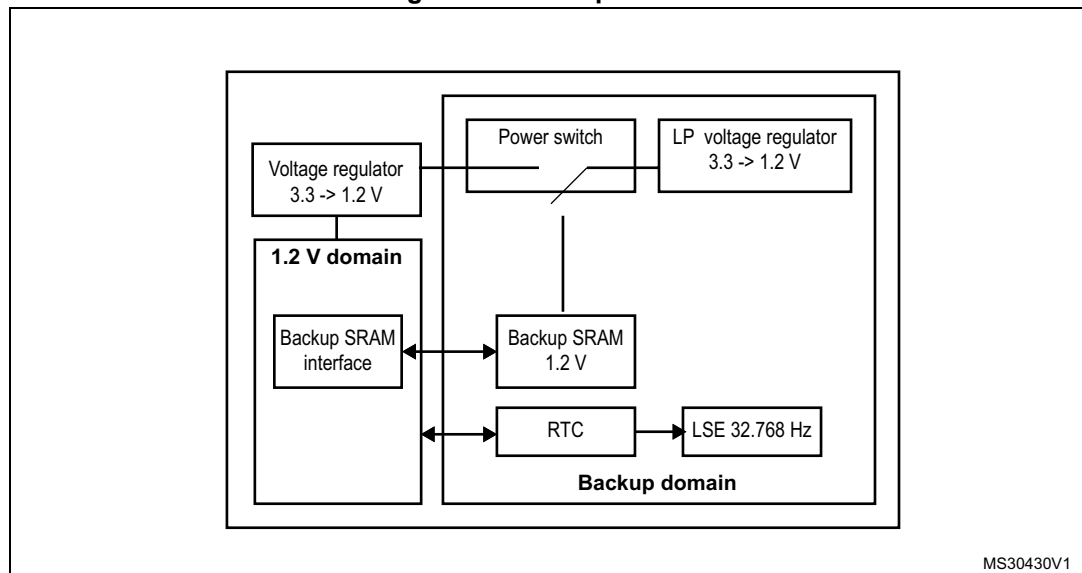
When the backup domain is supplied by V_{DD} (analog switch connected to V_{DD}), the backup SRAM is powered from V_{DD} which replaces the V_{BAT} power supply to save battery life.

When the backup domain is supplied by V_{BAT} (analog switch connected to V_{BAT} because V_{DD} is not present), the backup SRAM is powered by a dedicated low-power regulator. This regulator can be ON or OFF depending whether the application needs the backup SRAM function in Standby and V_{BAT} modes or not. The power-down of this regulator is controlled

by a dedicated bit, the BRE control bit of the PWR_CSR register (see [Section 5.4.2: PWR power control/status register \(PWR_CSR\) for STM32F405xx/07xx and STM32F415xx/17xx](#)).

The backup SRAM is not mass erased by a tamper event. When the Flash memory is read protected, the backup SRAM is also read protected to prevent confidential data, such as cryptographic private key, from being accessed. When the protection level change from level 1 to level 0 is requested, the backup SRAM content is erased.

Figure 11. Backup domain



5.1.3 Voltage regulator for STM32F405xx/07xx and STM32F415xx/17xx

An embedded linear voltage regulator supplies all the digital circuitries except for the backup domain and the Standby circuitry. The regulator output voltage is around 1.2 V.

This voltage regulator requires one or two external capacitors to be connected to one or two dedicated pins, V_{CAP_1} and V_{CAP_2} available in all packages. Specific pins must be connected either to V_{SS} or V_{DD} to activate or deactivate the voltage regulator. These pins depend on the package.

When activated by software, the voltage regulator is always enabled after Reset. It works in three different modes depending on the application modes.

- In **Run mode**, the regulator supplies full power to the 1.2 V domain (core, memories and digital peripherals). In this mode, the regulator output voltage (around 1.2 V) can be scaled by software to different voltage values:

Scale 1 or scale 2 can be configured on the fly through VOS[1:0] bits of the PWR_CR register.

The voltage scaling allows optimizing the power consumption when the device is clocked below the maximum system frequency.

- In **Stop mode**, the main regulator or the low-power regulator supplies to the 1.2 V domain, thus preserving the content of registers and internal SRAM. The voltage

regulator can be put either in main regulator mode (MR) or in low-power mode (LPR). The programmed voltage scale remains the same during Stop mode:

The programmed voltage scale remains the same during Stop mode (see [Section 5.4.1: PWR power control register \(PWR_CR\) for STM32F405xx/07xx and STM32F415xx/17xx](#)).

- In **Standby mode**, the regulator is powered down. The content of the registers and SRAM are lost except for the Standby circuitry and the backup domain.

Note: For more details, refer to the voltage regulator section in the STM32F405xx/07xx and STM32F415xx/17xx datasheets.

5.1.4 Voltage regulator for STM32F42xxx and STM32F43xxx

An embedded linear voltage regulator supplies all the digital circuitries except for the backup domain and the Standby circuitry. The regulator output voltage is around 1.2 V.

This voltage regulator requires two external capacitors to be connected to two dedicated pins, VCAP_1 and VCAP_2 available in all packages. Specific pins must be connected either to V_{SS} or V_{DD} to activate or deactivate the voltage regulator. These pins depend on the package.

When activated by software, the voltage regulator is always enabled after Reset. It works in three different modes depending on the application modes (Run, Stop, or Standby mode).

- In **Run mode**, the main regulator supplies full power to the 1.2 V domain (core, memories and digital peripherals). In this mode, the regulator output voltage (around 1.2 V) can be scaled by software to different voltage values (scale 1, scale 2, and scale 3 can be configured through VOS[1:0] bits of the PWR_CR register). The scale can be modified only when the PLL is OFF and the HSI or HSE clock source is selected as system clock source. The new value programmed is active only when the PLL is ON. When the PLL is OFF, the voltage scale 3 is automatically selected.

The voltage scaling allows optimizing the power consumption when the device is clocked below the maximum system frequency. After exit from Stop mode, the voltage

scale 3 is automatically selected. (see [Section 5.4.1: PWR power control register \(PWR_CR\) for STM32F405xx/07xx and STM32F415xx/17xx](#)).

2 operating modes are available:

- **Normal mode:** The CPU and core logic operate at maximum frequency at a given voltage scaling (scale 1, scale 2 or scale 3)
- **Over-drive mode:** This mode allows the CPU and the core logic to operate at a higher frequency than the normal mode for the voltage scaling scale 1 and scale 2.
- In **Stop mode:** the main regulator or low-power regulator supplies a low-power voltage to the 1.2V domain, thus preserving the content of registers and internal SRAM.
The voltage regulator can be put either in main regulator mode (MR) or in low-power mode (LPR). Both modes can be configured by software as follows:
 - **Normal mode:** the 1.2 V domain is preserved in nominal leakage mode. It is the default mode when the main regulator (MR) or the low-power regulator (LPR) is enabled.
 - **Under-drive mode:** the 1.2 V domain is preserved in reduced leakage mode. This mode is only available with the main regulator or in low-power regulator mode (see [Table 23](#)).
- In **Standby mode:** the regulator is powered down. The content of the registers and SRAM are lost except for the Standby circuitry and the backup domain.

Note: Over-drive and under-drive mode are not available when the regulator is bypassed.
For more details, refer to the voltage regulator section in the STM32F42xxx and STM32F43xxx datasheets.

Table 23. Voltage regulator configuration mode versus device operating mode⁽¹⁾

Voltage regulator configuration	Run mode	Sleep mode	Stop mode	Standby mode
Normal mode	MR	MR	MR or LPR	-
Over-drive mode ⁽²⁾	MR	MR	-	-
Under-drive mode	-	-	MR or LPR	-
Power-down mode	-	-	-	Yes

1. '-' means that the corresponding configuration is not available.
2. The over-drive mode is not available when $V_{DD} = 1.8$ to 2.1 V.

Entering Over-drive mode

It is recommended to enter Over-drive mode when the application is not running critical tasks and when the system clock source is either HSI or HSE. To optimize the configuration time, enable the Over-drive mode during the PLL lock phase.

To enter Over-drive mode, follow the sequence below:

1. Select HSI or HSE as system clock.
2. Configure RCC_PLLCFGR register and set PLLON bit of RCC_CR register.
3. Set ODEN bit of PWR_CR register to enable the Over-drive mode and wait for the ODRDY flag to be set in the PWR_CSR register.
4. Set the ODSW bit in the PWR_CR register to switch the voltage regulator from Normal mode to Over-drive mode. The System is stalled during the switch but the PLL clock system is still running during locking phase.
5. Wait for the ODSWRDY flag in the PWR_CSR to be set.
6. Select the required Flash latency as well as AHB and APB prescalers.
7. Wait for PLL lock.
8. Switch the system clock to the PLL.
9. Enable the peripherals that are not generated by the System PLL (I2S clock, LCD-TFT clock, SAI1 clock, USB_48MHz clock....).

Note: The PLLI2S and PLLSAI can be configured at the same time as the system PLL.

During the Over-drive switch activation, no peripheral clocks should be enabled. The peripheral clocks must be enabled once the Over-drive mode is activated.

Entering Stop mode disables the Over-drive mode, as well as the PLL. The application software has to configure again the Over-drive mode and the PLL after exiting from Stop mode.

Exiting from Over-drive mode

It is recommended to exit from Over-drive mode when the application is not running critical tasks and when the system clock source is either HSI or HSE. There are two sequences that allow exiting from over-drive mode:

- By resetting simultaneously the ODEN and ODSW bits in the PWR_CR register (sequence 1)
- By resetting first the ODSW bit to switch the voltage regulator to Normal mode and then resetting the ODEN bit to disable the Over-drive mode (sequence 2).

Example of sequence 1:

1. Select HSI or HSE as system clock source.
2. Disable the peripheral clocks that are not generated by the System PLL (I2S clock, LCD-TFT clock, SAI1 clock, USB_48MHz clock,...)
3. Reset simultaneously the ODEN and the ODSW bits in the PWR_CR register to switch back the voltage regulator to Normal mode and disable the Over-drive mode.
4. Wait for the ODWRDY flag of PWR_CSR to be reset.

Example of sequence 2:

1. Select HSI or HSE as system clock source.
2. Disable the peripheral clocks that are not generated by the System PLL (I2S clock, LCD-TFT clock, SAI1 clock, USB_48MHz clock,...).
3. Reset the ODSW bit in the PWR_CR register to switch back the voltage regulator to Normal mode. The system clock is stalled during voltage switching.
4. Wait for the ODWRDY flag of PWR_CSR to be reset.
5. Reset the ODEN bit in the PWR_CR register to disable the Over-drive mode.

Note: During step 3, the ODEN bit remains set and the Over-drive mode is still enabled but not active (ODSW bit is reset). If the ODEN bit is reset instead, the Over-drive mode is disabled and the voltage regulator is switched back to the initial voltage.

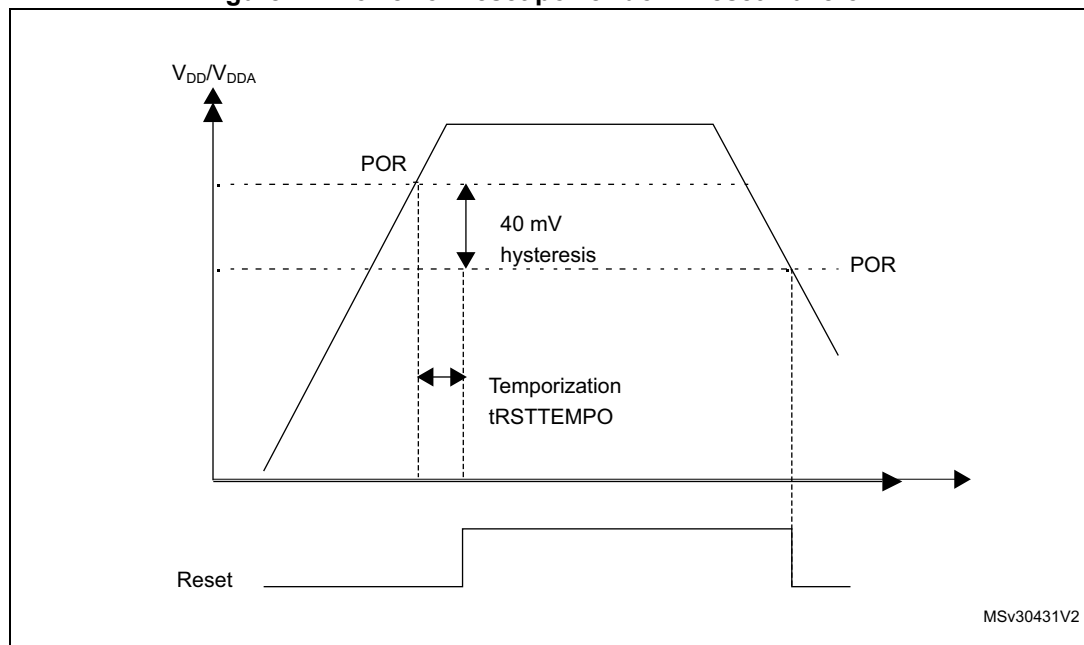
5.2 Power supply supervisor

5.2.1 Power-on reset (POR)/power-down reset (PDR)

The device has an integrated POR/PDR circuitry that allows proper operation starting from 1.8 V.

The device remains in Reset mode when V_{DD}/V_{DDA} is below a specified threshold, $V_{POR/PDR}$, without the need for an external reset circuit. For more details concerning the power on/power-down reset threshold, refer to the electrical characteristics of the datasheet.

Figure 12. Power-on reset/power-down reset waveform



5.2.2 Brownout reset (BOR)

During power on, the Brownout reset (BOR) keeps the device under reset until the supply voltage reaches the specified V_{BOR} threshold.

V_{BOR} is configured through device option bytes. By default, BOR is off. 3 programmable V_{BOR} threshold levels can be selected:

- BOR Level 3 (VBOR3). Brownout threshold level 3.
- BOR Level 2 (VBOR2). Brownout threshold level 2.
- BOR Level 1 (VBOR1). Brownout threshold level 1.

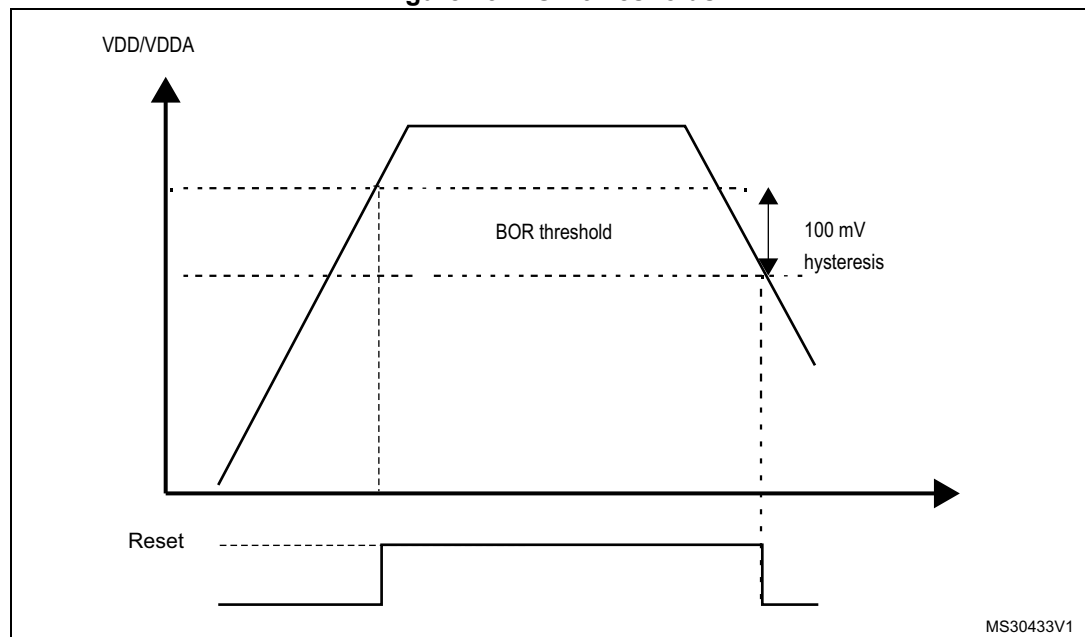
Note: For full details about BOR characteristics, refer to the "Electrical characteristics" section in the device datasheet.

When the supply voltage (V_{DD}) drops below the selected V_{BOR} threshold, a device reset is generated.

The BOR can be disabled by programming the device option bytes. In this case, the power-on and power-down is then monitored by the POR/ PDR (see [Section 5.2.1: Power-on reset \(POR\)/power-down reset \(PDR\)](#)).

The BOR threshold hysteresis is ~100 mV (between the rising and the falling edge of the supply voltage).

Figure 13. BOR thresholds



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5.2.3 Programmable voltage detector (PVD)

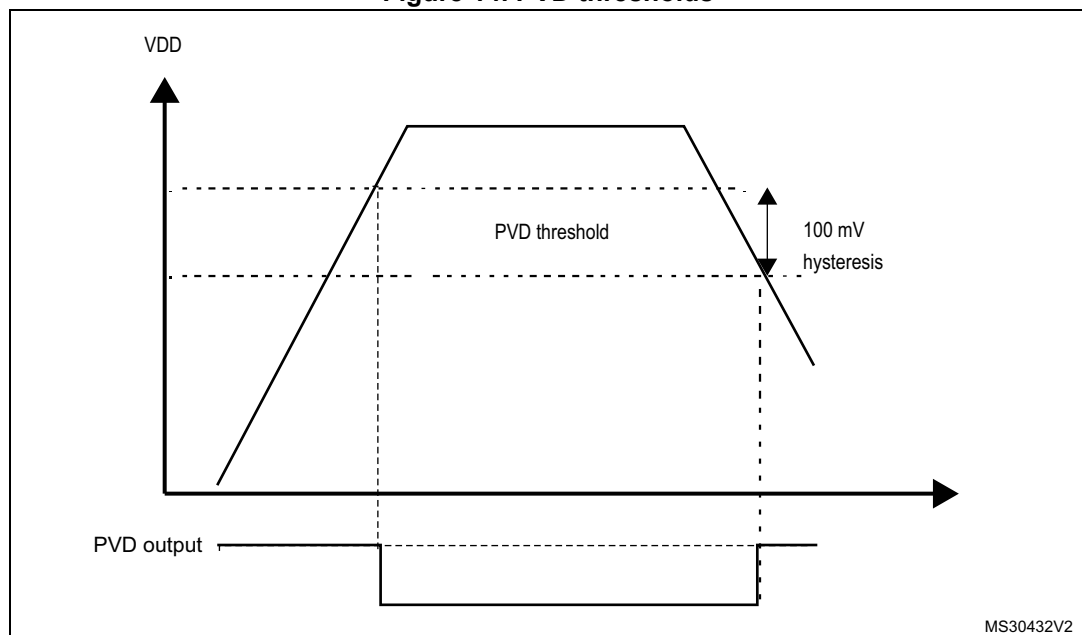
You can use the PVD to monitor the V_{DD} power supply by comparing it to a threshold selected by the PLS[2:0] bits in the [PWR power control register \(PWR_CR\)](#) for

STM32F405xx/07xx and STM32F415xx/17xx and PWR power control register (PWR_CR) for STM32F42xxx and STM32F43xxx.

The PVD is enabled by setting the PVDE bit.

A PVDO flag is available, in the *PWR power control/status register (PWR_CSR)* for *STM32F405xx/07xx and STM32F415xx/17xx*, to indicate if V_{DD} is higher or lower than the PVD threshold. This event is internally connected to the EXTI line 16 and can generate an interrupt if enabled through the EXTI registers. The PVD output interrupt can be generated when V_{DD} drops below the PVD threshold and/or when V_{DD} rises above the PVD threshold depending on EXTI line16 rising/falling edge configuration. As an example the service routine could perform emergency shutdown tasks.

Figure 14. PVD thresholds



5.3 Low-power modes

By default, the microcontroller is in Run mode after a system or a power-on reset. In Run mode the CPU is clocked by HCLK and the program code is executed. Several low-power modes are available to save power when the CPU does not need to be kept running, for example when waiting for an external event. It is up to the user to select the mode that gives the best compromise between low-power consumption, short startup time and available wake-up sources.

The devices feature three low-power modes:

- Sleep mode (Cortex[®]-M4 with FPU core stopped, peripherals kept running)
- Stop mode (all clocks are stopped)
- Standby mode (1.2 V domain powered off)

In addition, the power consumption in Run mode can be reduced by one of the following means:

- Slowing down the system clocks
- Gating the clocks to the APBx and AHBx peripherals when they are unused.

Entering low-power mode

Low-power modes are entered by the MCU by executing the WFI (Wait For Interrupt), or WFE (Wait for Event) instructions, or when the SLEEPONEXIT bit in the Cortex®-M4 with FPU System Control register is set on Return from ISR.

Entering Low-power mode through WFI or WFE is executed only if no interrupt is pending or no event is pending.

Exiting low-power mode

The MCU exits from Sleep and Stop modes low-power mode depending on the way the low-power mode was entered:

- If the WFI instruction or Return from ISR was used to enter the low-power mode, any peripheral interrupt acknowledged by the NVIC can wake up the device.
- If the WFE instruction is used to enter the low-power mode, the MCU exits the low-power mode as soon as an event occurs. The wake-up event can be generated either by:
 - NVIC IRQ interrupt:

When SEVONPEND = 0 in the Cortex®-M4 with FPU System Control register: by enabling an interrupt in the peripheral control register and in the NVIC. When the MCU resumes from WFE, the peripheral interrupt pending bit and the NVIC peripheral IRQ channel pending bit (in the NVIC interrupt clear pending register) have to be cleared. Only NVIC interrupts with sufficient priority wakes up and interrupts the MCU.

When SEVONPEND = 1 in the Cortex®-M4 with FPU System Control register: by enabling an interrupt in the peripheral control register and optionally in the NVIC. When the MCU resumes from WFE, the peripheral interrupt pending bit and when enabled the NVIC peripheral IRQ channel pending bit (in the NVIC interrupt clear pending register) have to be cleared. All NVIC interrupts wakes up the MCU, even the disabled ones. Only enabled NVIC interrupts with sufficient priority wakes up and interrupts the MCU.
 - Event

This is done by configuring a EXTI line in event mode. When the CPU resumes from WFE, it is not necessary to clear the EXTI peripheral interrupt pending bit or the NVIC IRQ channel pending bit as the pending bits corresponding to the event line is not set. It may be necessary to clear the interrupt flag in the peripheral.

The MCU exits from Standby low-power mode through an external reset (NRST pin), an IWDG reset, a rising edge on one of the enabled WKUPx pins or a RTC event occurs (see [Figure 237: RTC block diagram](#)).

After waking up from Standby mode, program execution restarts in the same way as after a Reset (boot pin sampling, option bytes loading, reset vector is fetched, etc.).

Only enabled NVIC interrupts with sufficient priority wakes up and interrupts the MCU.

Table 24. Low-power mode summary

Mode name	Entry	Wake-up	Effect on 1.2 V domain clocks	Effect on V _{DD} domain clocks	Voltage regulator
Sleep (Sleep now or Sleep-on-exit)	WFI or Return from ISR	Any interrupt	CPU CLK OFF no effect on other clocks or analog clock sources	None	ON
	WFE	Wake-up event			
Stop	PDDS and LPDS bits + SLEEPDEEP bit + WFI, Return from ISR or WFE	Any EXTI line (configured in the EXTI registers, internal and external lines)	All 1.2 V domain clocks OFF	HSI and HSE oscillators OFF	ON or in low-power mode (depends on PWR power control register (PWR_CR) for STM32F405xx/07xx and STM32F415xx/17xx and PWR power control register (PWR_CR) for STM32F405xx/07xx and STM32F415xx/17xx PWR power control register (PWR_CR) for STM32F42xxx and STM32F43xxx)
Standby	PDDS bit + SLEEPDEEP bit + WFI, Return from ISR or WFE	WKUP pin rising edge, RTC alarm (Alarm A or Alarm B), RTC Wake-up event, RTC tamper events, RTC time stamp event, external reset in NRST pin, IWDG reset			OFF

5.3.1 Slowing down system clocks

In Run mode the speed of the system clocks (SYSCLK, HCLK, PCLK1, PCLK2) can be reduced by programming the prescaler registers. These prescalers can also be used to slow down peripherals before entering Sleep mode.

For more details refer to [Section 7.3.3: RCC clock configuration register \(RCC_CFGR\)](#).

5.3.2 Peripheral clock gating

In Run mode, the HCLKx and PCLKx for individual peripherals and memories can be stopped at any time to reduce power consumption.

To further reduce power consumption in Sleep mode the peripheral clocks can be disabled prior to executing the WFI or WFE instructions.

Peripheral clock gating is controlled by the AHB1 peripheral clock enable register (RCC_AHB1ENR), AHB2 peripheral clock enable register (RCC_AHB2ENR), AHB3 peripheral clock enable register (RCC_AHB3ENR) (see [Section 7.3.10: RCC AHB1 peripheral clock enable register \(RCC_AHB1ENR\)](#), [Section 7.3.11: RCC AHB2 peripheral clock enable register \(RCC_AHB2ENR\)](#), [Section 7.3.12: RCC AHB3 peripheral clock enable register \(RCC_AHB3ENR\)](#) for STM32F405xx/07xx and STM32F415xx/17xx, and [Section 6.3.10: RCC AHB1 peripheral clock register \(RCC_AHB1ENR\)](#), [Section 6.3.11: RCC AHB2 peripheral clock enable register \(RCC_AHB2ENR\)](#), and [Section 6.3.12: RCC AHB3 peripheral clock enable register \(RCC_AHB3ENR\)](#) for STM32F42xxx and STM32F43xxx).

Disabling the peripherals clocks in Sleep mode can be performed automatically by resetting the corresponding bit in RCC_AHBxLPENR and RCC_APBxLPENR registers.

5.3.3 Sleep mode

Entering Sleep mode

The Sleep mode is entered according to [Section : Entering low-power mode](#), when the SLEEPDEEP bit in the Cortex®-M4 with FPU System Control register is cleared.

Refer to [Table 25](#) and [Table 26](#) for details on how to enter Sleep mode.

Note: All interrupt pending bits must be cleared before the sleep mode entry.

Exiting Sleep mode

The Sleep mode is exited according to [Section : Exiting low-power mode](#).

Refer to [Table 25](#) and [Table 26](#) for more details on how to exit Sleep mode.

Table 25. Sleep-now entry and exit

Sleep-now mode	Description
Mode entry	WFI (Wait for Interrupt) or WFE (Wait for Event) while: – SLEEPDEEP = 0, and – No interrupt (for WFI) or event (for WFE) is pending. Refer to the Cortex®-M4 with FPU System Control register.
	On Return from ISR while: – SLEEPDEEP = 0 and – SLEEPONEXIT = 1, – No interrupt is pending. Refer to the Cortex®-M4 with FPU System Control register.

Table 25. Sleep-now entry and exit (continued)

Sleep-now mode	Description
Mode exit	If WFI or Return from ISR was used for entry: Interrupt: Refer to Table 62: Vector table for STM32F405xx/07xx and STM32F415xx/17xx and Table 63: Vector table for STM32F42xxx and STM32F43xxx If WFE was used for entry and SEVONPEND = 0 Wake-up event: Refer to Section 12.2.3: Wake-up event management If WFE was used for entry and SEVONPEND = 1 Interrupt even when disabled in NVIC: refer to Table 62: Vector table for STM32F405xx/07xx and STM32F415xx/17xx and Table 63: Vector table for STM32F42xxx and STM32F43xxx or Wake-up event (see Section 12.2.3: Wake-up event management).
Wake-up latency	None

Table 26. Sleep-on-exit entry and exit

Sleep-on-exit	Description
Mode entry	WFI (Wait for Interrupt) or WFE (Wait for Event) while: – SLEEPDEEP = 0, and – No interrupt (for WFI) or event (for WFE) is pending. Refer to the Cortex[®]-M4 with FPU System Control register. On Return from ISR while: – SLEEPDEEP = 0, and – SLEEPONEXIT = 1, and – No interrupt is pending. Refer to the Cortex[®]-M4 with FPU System Control register.
Mode exit	Interrupt: refer to Table 62: Vector table for STM32F405xx/07xx and STM32F415xx/17xx and Table 63: Vector table for STM32F42xxx and STM32F43xxx
Wake-up latency	None

5.3.4 Stop mode (STM32F405xx/07xx and STM32F415xx/17xx)

The Stop mode is based on the Cortex[®]-M4 with FPU deepsleep mode combined with peripheral clock gating. The voltage regulator can be configured either in normal or low-power mode. In Stop mode, all clocks in the 1.2 V domain are stopped, the PLLs, the HSI and the HSE RC oscillators are disabled. Internal SRAM and register contents are preserved.

By setting the FPDS bit in the PWR_CR register, the Flash memory also enters power-down mode when the device enters Stop mode. When the Flash memory is in power-down mode, an additional startup delay is incurred when waking up from Stop mode (see [Table 27: Stop operating modes \(STM32F405xx/07xx and STM32F415xx/17xx\)](#) and [Section 5.4.1: PWR power control register \(PWR_CR\) for STM32F405xx/07xx and STM32F415xx/17xx](#)).

**Table 27. Stop operating modes
(STM32F405xx/07xx and STM32F415xx/17xx)**

Stop mode	LPDS bit	FPDS bit	Wake-up latency
STOP MR (Main regulator)	0	0	HSI RC startup time
STOP MR-FPD	0	1	HSI RC startup time + Flash wake-up time from Power Down mode
STOP LP	1	0	HSI RC startup time + regulator wake-up time from LP mode
STOP LP-FPD	1	1	HSI RC startup time + Flash wake-up time from Power Down mode + regulator wake-up time from LP mode

Entering Stop mode (for STM32F405xx/07xx and STM32F415xx/17xx)

The Stop mode is entered according to [Section : Entering low-power mode](#), when the SLEEPDEEP bit in the Cortex[®]-M4 with FPU System Control register is set.

Refer to [Table 28](#) for details on how to enter the Stop mode.

To further reduce power consumption in Stop mode, the internal voltage regulator can be put in low-power mode. This is configured by the LPDS bit of the [PWR power control register \(PWR_CR\) for STM32F405xx/07xx and STM32F415xx/17xx](#) and [PWR power control register \(PWR_CR\) for STM32F42xxx and STM32F43xxx](#).

If Flash memory programming is ongoing, the Stop mode entry is delayed until the memory access is finished.

If an access to the APB domain is ongoing, The Stop mode entry is delayed until the APB access is finished.

In Stop mode, the following features can be selected by programming individual control bits:

- Independent watchdog (IWDG): the IWDG is started by writing to its Key register or by hardware option. Once started it cannot be stopped except by a Reset. See [Section 21.3](#) in [Section 21: Independent watchdog \(IWDG\)](#).
- Real-time clock (RTC): this is configured by the RTCEN bit in the [Section 7.3.20: RCC Backup domain control register \(RCC_BDCR\)](#)
- Internal RC oscillator (LSI RC): this is configured by the LSION bit in the [Section 7.3.21: RCC clock control & status register \(RCC_CSR\)](#).
- External 32.768 kHz oscillator (LSE OSC): this is configured by the LSEON bit in the [Section 7.3.20: RCC Backup domain control register \(RCC_BDCR\)](#).

The ADC or DAC can also consume power during the Stop mode, unless they are disabled before entering it. To disable them, the ADON bit in the ADC_CR2 register and the ENx bit in the DAC_CR register must both be written to 0.

Note: *If the application needs to disable the external clock before entering Stop mode, the HSEON bit must first be disabled and the system clock switched to HSI.*

Otherwise, if the HSEON bit is kept enabled while the external clock (external oscillator) can be removed before entering stop mode, the clock security system (CSS) feature must be enabled to detect any external oscillator failure and avoid a malfunction behavior when entering stop mode.

Exiting Stop mode (for STM32F405xx/07xx and STM32F415xx/17xx)

The Stop mode is exited according to [Section : Exiting low-power mode](#).

Refer to [Table 28](#) for more details on how to exit Stop mode.

When exiting Stop mode by issuing an interrupt or a wake-up event, the HSI RC oscillator is selected as system clock.

When the voltage regulator operates in low-power mode, an additional startup delay is incurred when waking up from Stop mode. By keeping the internal regulator ON during Stop mode, the consumption is higher although the startup time is reduced.

Table 28. Stop mode entry and exit (for STM32F405xx/07xx and STM32F415xx/17xx)

Stop mode	Description
Mode entry	WFI (Wait for Interrupt) or WFE (Wait for Event) while: – No interrupt (for WFI) or event (for WFE) is pending, – SLEEPDEEP bit is set in Cortex®-M4 with FPU System Control register, – PDDS bit is cleared in Power Control register (PWR_CR), – Select the voltage regulator mode by configuring LPDS bit in PWR_CR.
	On Return from ISR: – No interrupt is pending, – SLEEPDEEP bit is set in Cortex®-M4 with FPU System Control register, – SLEEPONEXIT = 1, – PDDS bit is cleared in Power Control register (PWR_CR).
	Note: <i>To enter Stop mode, all EXTI Line pending bits (in Pending register (EXTI_PR)), all peripheral interrupts pending bits, the RTC Alarm (Alarm A and Alarm B), RTC wake-up, RTC tamper, and RTC time stamp flags, must be reset. Otherwise, the Stop mode entry procedure is ignored and program execution continues.</i>

Table 28. Stop mode entry and exit (for STM32F405xx/07xx and STM32F415xx/17xx)

Stop mode	Description
Mode exit	If WFI or Return from ISR was used for entry: Any EXTI lines configured in Interrupt mode (the corresponding EXTI Interrupt vector must be enabled in the NVIC). The interrupt source can be external interrupts or peripherals with wake-up capability. Refer to Table 62: Vector table for STM32F405xx/07xx and STM32F415xx/17xx on page 374 and Table 63: Vector table for STM32F42xxx and STM32F43xxx. If WFE was used for entry and SEVONPEND = 0 Any EXTI lines configured in event mode. Refer to Section 12.2.3: Wake-up event management on page 382. If WFE was used for entry and SEVONPEND = 1: - Any EXTI lines configured in Interrupt mode (even if the corresponding EXTI Interrupt vector is disabled in the NVIC). The interrupt source can be an external interrupt or a peripheral with wake-up capability. Refer to Table 62: Vector table for STM32F405xx/07xx and STM32F415xx/17xx on page 374 and Table 63: Vector table for STM32F42xxx and STM32F43xxx. - Wake-up event: refer to Section 12.2.3: Wake-up event management on page 382.
Wake-up latency	Table 27: Stop operating modes (STM32F405xx/07xx and STM32F415xx/17xx)

5.3.5 Stop mode (STM32F42xxx and STM32F43xxx)

The Stop mode is based on the Cortex[®]-M4 with FPU deepsleep mode combined with peripheral clock gating. The voltage regulator can be configured either in normal or low-power mode. In Stop mode, all clocks in the 1.2 V domain are stopped, the PLLs, the HSI and the HSE RC oscillators are disabled. Internal SRAM and register contents are preserved.

In Stop mode, the power consumption can be further reduced by using additional settings in the PWR_CR register. However this induces an additional startup delay when waking up from Stop mode (see [Table 29](#)).

Table 29. Stop operating modes (STM32F42xxx and STM32F43xxx)

Voltage Regulator Mode		UDEN[1:0] bits	MRUDS bit	LPUDS bit	LPDS bit	FPDS bit	Wake-up latency
Normal mode	STOP MR (Main Regulator)	-	0	-	0	0	HSI RC startup time
	STOP MR- FPD	-	0	-	0	1	HSI RC startup time + Flash wake-up time from power-down mode
	STOP LP	-	0	0	1	0	HSI RC startup time + regulator wake-up time from LP mode
	STOP LP-FPD	-	-	0	1	1	HSI RC startup time + Flash wake-up time from power-down mode + regulator wake-up time from LP mode
Under-drive Mode	STOP UMR-FPD	3	1	-	0	-	HSI RC startup time + Flash wake-up time from power-down mode + Main regulator wake-up time from under-drive mode + Core logic to nominal mode
	STOP ULP-FPD	3	-	1	1	-	HSI RC startup time + Flash wake-up time from power-down mode + regulator wake-up time from LP under-drive mode + Core logic to nominal mode

Entering Stop mode (STM32F42xxx and STM32F43xxx)

The Stop mode is entered according to [Section : Entering low-power mode](#), when the SLEEPDEEP bit in the Cortex[®]-M4 with FPU System Control register is set.

Refer to [Table 30](#) for details on how to enter the Stop mode.

When the microcontroller enters in Stop mode, the voltage scale 3 is automatically selected. To further reduce power consumption in Stop mode, the internal voltage regulator can be put in low voltage mode. This is configured by the LPDS, MRUDS, LPUDS and UDEN bits of the [PWR power control register \(PWR_CR\) for STM32F405xx/07xx and STM32F415xx/17xx](#).

If Flash memory programming is ongoing, the Stop mode entry is delayed until the memory access is finished.

If an access to the APB domain is ongoing, The Stop mode entry is delayed until the APB access is finished.

If the Over-drive mode was enabled before entering Stop mode, it is automatically disabled during when the Stop mode is activated.

In Stop mode, the following features can be selected by programming individual control bits:

- Independent watchdog (IWDG): the IWDG is started by writing to its Key register or by hardware option. Once started it cannot be stopped except by a Reset. See [Section 21.3](#) in [Section 21: Independent watchdog \(IWDG\)](#).
- Real-time clock (RTC): this is configured by the RTCEN bit in the [Section 7.3.20: RCC Backup domain control register \(RCC_BDCR\)](#)
- Internal RC oscillator (LSI RC): this is configured by the LSION bit in the [Section 7.3.21: RCC clock control & status register \(RCC_CSR\)](#).
- External 32.768 kHz oscillator (LSE OSC): this is configured by the LSEON bit in the [RCC Backup domain control register \(RCC_BDCR\)](#).

The ADC or DAC can also consume power during the Stop mode, unless they are disabled before entering it. To disable them, the ADON bit in the ADC_CR2 register and the ENx bit in the DAC_CR register must both be written to 0.

Note: Before entering Stop mode, it is recommended to enable the clock security system (CSS) feature to prevent external oscillator (HSE) failure from impacting the internal MCU behavior.

Exiting Stop mode (STM32F42xxx and STM32F43xxx)

The Stop mode is exited according to [Section : Exiting low-power mode](#).

Refer to [Table 30](#) for more details on how to exit Stop mode.

When exiting Stop mode by issuing an interrupt or a wake-up event, the HSI RC oscillator is selected as system clock.

If the Under-drive mode was enabled, it is automatically disabled after exiting Stop mode.

When the voltage regulator operates in low voltage mode, an additional startup delay is incurred when waking up from Stop mode. By keeping the internal regulator ON during Stop mode, the consumption is higher although the startup time is reduced.

When the voltage regulator operates in Under-drive mode, an additional startup delay is induced when waking up from Stop mode.

Table 30. Stop mode entry and exit (STM32F42xxx and STM32F43xxx)

Stop mode	Description
Mode entry	WFI (Wait for Interrupt) or WFE (Wait for Event) while: – No interrupt or event is pending, – SLEEPDEEP bit is set in Cortex[®]-M4 with FPU System Control register, – PDDS bit is cleared in Power Control register (PWR_CR), – Select the voltage regulator mode by configuring LPDS, MRUDS, LPUDS and UDEN bits in PWR_CR (see Table 29: Stop operating modes (STM32F42xxx and STM32F43xxx)).
	On Return from ISR while: – No interrupt is pending, – SLEEPDEEP bit is set in Cortex[®]-M4 with FPU System Control register, and – SLEEPONEXIT = 1, and – PDDS is cleared in PWR_CR1.
	<i>Note: To enter Stop mode, all EXTI Line pending bits (in Pending register (EXTI_PR)), all peripheral interrupts pending bits, the RTC Alarm (Alarm A and Alarm B), RTC wake-up, RTC tamper, and RTC time stamp flags, must be reset. Otherwise, the Stop mode entry procedure is ignored and program execution continues.</i>
Mode exit	If WFI or Return from ISR was used for entry: All EXTI lines configured in Interrupt mode (the corresponding EXTI Interrupt vector must be enabled in the NVIC). The interrupt source can be external interrupts or peripherals with wake-up capability. Refer to Table 62: Vector table for STM32F405xx/07xx and STM32F415xx/17xx on page 374. If WFE was used for entry and SEVONPEND = 0: All EXTI Lines configured in event mode. Refer to Section 12.2.3: Wake-up event management on page 382 If WFE was used for entry and SEVONPEND = 1: – Any EXTI lines configured in Interrupt mode (even if the corresponding EXTI Interrupt vector is disabled in the NVIC). The interrupt source can be external interrupts or peripherals with wake-up capability. Refer to Table 62: Vector table for STM32F405xx/07xx and STM32F415xx/17xx on page 374 and Table 63: Vector table for STM32F42xxx and STM32F43xxx. – Wake-up event: refer to Section 12.2.3: Wake-up event management on page 382.
Wake-up latency	Refer to Table 29: Stop operating modes (STM32F42xxx and STM32F43xxx)

5.3.6 Standby mode

The Standby mode allows to achieve the lowest power consumption. It is based on the Cortex[®]-M4 with FPU deepsleep mode, with the voltage regulator disabled. The 1.2 V domain is consequently powered off. The PLLs, the HSI oscillator and the HSE oscillator are also switched off. SRAM and register contents are lost except for registers in the backup domain (RTC registers, RTC backup register and backup SRAM), and Standby circuitry (see [Figure 9](#)).

Entering Standby mode

The Standby mode is entered according to [Section : Entering low-power mode](#), when the SLEEPDEEP bit in the Cortex[®]-M4 with FPU System Control register is set.

Refer to [Table 31](#) for more details on how to enter Standby mode.

In Standby mode, the following features can be selected by programming individual control bits:

- Independent watchdog (IWDG): the IWDG is started by writing to its Key register or by hardware option. Once started it cannot be stopped except by a reset. See [Section 21.3](#) in [Section 21: Independent watchdog \(IWDG\)](#).
- Real-time clock (RTC): this is configured by the RTCEN bit in the backup domain control register (RCC_BDCR)
- Internal RC oscillator (LSI RC): this is configured by the LSION bit in the Control/status register (RCC_CSR).
- External 32.768 kHz oscillator (LSE OSC): this is configured by the LSEON bit in the backup domain control register (RCC_BDCR)

Exiting Standby mode

The Standby mode is exited according to [Section : Exiting low-power mode](#). The SBF status flag in PWR_CR (see [Section 5.4.2: PWR power control/status register \(PWR_CSR\) for STM32F405xx/07xx and STM32F415xx/17xx](#)) indicates that the MCU was in Standby mode. All registers are reset after wake-up from Standby except for PWR_CR.

Refer to [Table 31](#) for more details on how to exit Standby mode.

Table 31. Standby mode entry and exit

Standby mode	Description
Mode entry	WFI (Wait for Interrupt) or WFE (Wait for Event) while: – SLEEPDEEP is set in Cortex[®]-M4 with FPU System Control register, – PDDS bit is set in Power Control register (PWR_CR), – No interrupt (for WFI) or event (for WFE) is pending, – WUF bit is cleared in Power Control register (PWR_CR), – the RTC flag corresponding to the chosen wake-up source (RTC Alarm A, RTC Alarm B, RTC wake-up, Tamper or Timestamp flags) is cleared
	On return from ISR while: – SLEEPDEEP bit is set in Cortex[®]-M4 with FPU System Control register, and – SLEEPONEXIT = 1, and – PDDS bit is set in Power Control register (PWR_CR), and – No interrupt is pending, – WUF bit is cleared in Power Control/Status register (PWR_SR), – The RTC flag corresponding to the chosen wake-up source (RTC Alarm A, RTC Alarm B, RTC wake-up, Tamper or Timestamp flags) is cleared.
Mode exit	WKUP pin rising edge, RTC alarm (Alarm A and Alarm B), RTC wake-up, tamper event, time stamp event, external reset in NRST pin, IWDG reset.
Wake-up latency	Reset phase.

I/O states in Standby mode

In Standby mode, all I/O pins are high impedance except for:

- Reset pad (still available)
- RTC_AF1 pin (PC13) if configured for tamper, time stamp, RTC Alarm out, or RTC clock calibration out
- WKUP pin (PA0), if enabled

Debug mode

By default, the debug connection is lost if the application puts the MCU in Stop or Standby mode while the debug features are used. This is due to the fact that the Cortex®-M4 with FPU core is no longer clocked.

However, by setting some configuration bits in the DBGMCU_CR register, the software can be debugged even when using the low-power modes extensively. For more details, refer to [Section 38.16.1: Debug support for low-power modes](#).

5.3.7 Programming the RTC alternate functions to wake up the device from the Stop and Standby modes

The MCU can be woken up from a low-power mode by an RTC alternate function.

The RTC alternate functions are the RTC alarms (Alarm A and Alarm B), RTC wake-up, RTC tamper event detection and RTC time stamp event detection.

These RTC alternate functions can wake up the system from the Stop and Standby low-power modes.

The system can also wake up from low-power modes without depending on an external interrupt (Auto-wake-up mode), by using the RTC alarm or the RTC wake-up events.

The RTC provides a programmable time base for waking up from the Stop or Standby mode at regular intervals.

For this purpose, two of the three alternate RTC clock sources can be selected by programming the RTCSEL[1:0] bits in the [Section 7.3.20: RCC Backup domain control register \(RCC_BDCR\)](#):

- Low-power 32.768 kHz external crystal oscillator (LSE OSC)
This clock source provides a precise time base with a very low-power consumption (additional consumption of less than 1 µA under typical conditions)
- Low-power internal RC oscillator (LSI RC)
This clock source has the advantage of saving the cost of the 32.768 kHz crystal. This internal RC oscillator is designed to use minimum power.

RTC alternate functions to wake up the device from the Stop mode

- To wake up the device from the Stop mode with an RTC alarm event, it is necessary to:
 - a) Configure the EXTI Line 17 to be sensitive to rising edges (Interrupt or Event modes)
 - b) Enable the RTC Alarm Interrupt in the RTC_CR register
 - c) Configure the RTC to generate the RTC alarm
- To wake up the device from the Stop mode with an RTC tamper or time stamp event, it is necessary to:
 - a) Configure the EXTI Line 21 to be sensitive to rising edges (Interrupt or Event modes)
 - b) Enable the RTC time stamp Interrupt in the RTC_CR register or the RTC tamper interrupt in the RTC_TAFCR register
 - c) Configure the RTC to detect the tamper or time stamp event
- To wake up the device from the Stop mode with an RTC wake-up event, it is necessary to:
 - a) Configure the EXTI Line 22 to be sensitive to rising edges (Interrupt or Event modes)
 - b) Enable the RTC wake-up interrupt in the RTC_CR register
 - c) Configure the RTC to generate the RTC Wake-up event

RTC alternate functions to wake up the device from the Standby mode

- To wake up the device from the Standby mode with an RTC alarm event, it is necessary to:
 - a) Enable the RTC alarm interrupt in the RTC_CR register
 - b) Configure the RTC to generate the RTC alarm
- To wake up the device from the Standby mode with an RTC tamper or time stamp event, it is necessary to:
 - a) Enable the RTC time stamp interrupt in the RTC_CR register or the RTC tamper interrupt in the RTC_TAFCR register
 - b) Configure the RTC to detect the tamper or time stamp event
- To wake up the device from the Standby mode with an RTC wake-up event, it is necessary to:
 - a) Enable the RTC wake-up interrupt in the RTC_CR register
 - b) Configure the RTC to generate the RTC wake-up event

Safe RTC alternate function wake-up flag clearing sequence

If the selected RTC alternate function is set before the PWR wake-up flag (WUTF) is cleared, it is not detected on the next event as detection is made once on the rising edge.

To avoid bouncing on the pins onto which the RTC alternate functions are mapped, and exit correctly from the Stop and Standby modes, it is recommended to follow the sequence below before entering the Standby mode:

- When using RTC alarm to wake up the device from the low-power modes:
 - a) Disable the RTC alarm interrupt (ALRAIE or ALRBIE bits in the RTC_CR register)
 - b) Clear the RTC alarm (ALRAF/ALRBF) flag

- c) Clear the PWR Wake-up (WUF) flag
- d) Enable the RTC alarm interrupt
- e) Re-enter the low-power mode
- When using RTC wake-up to wake up the device from the low-power modes:
 - a) Disable the RTC Wake-up interrupt (WUTIE bit in the RTC_CR register)
 - b) Clear the RTC Wake-up (WUTF) flag
 - c) Clear the PWR Wake-up (WUF) flag
 - d) Enable the RTC Wake-up interrupt
 - e) Re-enter the low-power mode
- When using RTC tamper to wake up the device from the low-power modes:
 - a) Disable the RTC tamper interrupt (TAMPIE bit in the RTC_TAFCR register)
 - b) Clear the Tamper (TAMP1F/TSF) flag
 - c) Clear the PWR Wake-up (WUF) flag
 - d) Enable the RTC tamper interrupt
 - e) Re-enter the low-power mode
- When using RTC time stamp to wake up the device from the low-power modes:
 - a) Disable the RTC time stamp interrupt (TSIE bit in RTC_CR)
 - b) Clear the RTC time stamp (TSF) flag
 - c) Clear the PWR Wake-up (WUF) flag
 - d) Enable the RTC TimeStamp interrupt
 - e) Re-enter the low-power mode

5.4 Power control registers (STM32F405xx/07xx and STM32F415xx/17xx)

5.4.1 PWR power control register (PWR_CR) for STM32F405xx/07xx and STM32F415xx/17xx

Address offset: 0x00

Reset value: 0x0000 4000 (reset by wake-up from Standby mode)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	VOS	Reserved				FPDS	DBP	PLS[2:0]			PVDE	CSBF	CWUF	PDDS	LPDS
	rw					rw	rw	rw	rw	rw	rw	w	w	rw	rw

Bits 31:15 Reserved, must be kept at reset value.

Bit 14 **VOS**: Regulator voltage scaling output selection

This bit controls the main internal voltage regulator output voltage to achieve a trade-off between performance and power consumption when the device does not operate at the maximum frequency.

0: Scale 2 mode

1: Scale 1 mode (default value at reset)

Bits 13:10 Reserved, must be kept at reset value.

Bit 9 **FPDS**: Flash power-down in Stop mode

When set, the Flash memory enters power-down mode when the device enters Stop mode. This allows to achieve a lower consumption in stop mode but a longer restart time.

0: Flash memory not in power-down when the device is in Stop mode

1: Flash memory in power-down when the device is in Stop mode

Bit 8 **DBP**: Disable backup domain write protection

In reset state, the RCC_BDCR register, the RTC registers (including the backup registers), and the BRE bit of the PWR_CSR register, are protected against parasitic write access. This bit must be set to enable write access to these registers.

0: Access to RTC and RTC Backup registers and backup SRAM disabled

1: Access to RTC and RTC Backup registers and backup SRAM enabled

Note: Depending on the APB1 prescaler, there is a delay between writing to DBP and the effective disabling/enabling of the backup domain protection. Therefore, a dummy read operation to the PWR_CR register is required just after writing to the DBP bit.

Bits 7:5 **PLS[2:0]**: PVD level selection

These bits are written by software to select the voltage threshold detected by the programmable voltage detector

000: 2.0 V

001: 2.1 V

010: 2.3 V

011: 2.5 V

100: 2.6 V

101: 2.7 V

110: 2.8 V

111: 2.9 V

Note: Refer to the electrical characteristics of the datasheet for more details.

Bit 4 **PVDE**: Programmable voltage detector enable

This bit is set and cleared by software.

0: PVD disabled

1: PVD enabled

Bit 3 **CSBF**: Clear standby flag

This bit is always read as 0.

0: No effect

1: Clear the SBF Standby Flag (write).

Bit 2 **CWUF**: Clear wake-up flag

This bit is always read as 0.

0: No effect

1: Clear the WUF Wake-up Flag **after 2 System clock cycles**

Bit 1 **PDDS**: Power-down deepsleep

This bit is set and cleared by software. It works together with the LPDS bit.

0: Enter Stop mode when the CPU enters deepsleep. The regulator status depends on the LPDS bit.

1: Enter Standby mode when the CPU enters deepsleep.

Bit 0 **LPDS**: Low-power deepsleep

This bit is set and cleared by software. It works together with the PDDS bit.

0: Voltage regulator on during Stop mode

1: Voltage regulator in low-power mode during Stop mode

5.4.2 PWR power control/status register (PWR_CSR) for STM32F405xx/07xx and STM32F415xx/17xx

Address offset: 0x04

Reset value: 0x0000 0000 (not reset by wake-up from Standby mode)

Additional APB cycles are needed to read this register versus a standard APB read.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res	VOS RDY	Reserved				BRE	EWUP	Reserved				BRR	PVDO	SBF	WUF
	r					rw	rw					r	r	r	r

Bits 31:15 Reserved, must be kept at reset value.

Bit 14 **VOSRDY**: Regulator voltage scaling output selection ready bit

0: Not ready

1: Ready

Bits 13:10 Reserved, must be kept at reset value.

Bit 9 **BRE**: Backup regulator enable

When set, the Backup regulator (used to maintain backup SRAM content in Standby and V_{BAT} modes) is enabled. If BRE is reset, the backup regulator is switched off. The backup SRAM can still be used but its content is lost in the Standby and V_{BAT} modes. Once set, the application must wait that the Backup Regulator Ready flag (BRR) is set to indicate that the data written into the RAM is maintained in the Standby and V_{BAT} modes.

0: Backup regulator disabled

1: Backup regulator enabled

Note: This bit is not reset when the device wakes up from Standby mode, by a system reset, or by a power reset. This bit is reset by a backup domain reset.

The DBP bit of the PWR_CR register must be set before BRE can be written.

Bit 8 **EWUP**: Enable WKUP pin

This bit is set and cleared by software.

0: WKUP pin is used for general purpose I/O. An event on the WKUP pin does not wake up the device from Standby mode.

1: WKUP pin is used for wake-up from Standby mode and forced in input pull down configuration (rising edge on WKUP pin wakes-up the system from Standby mode).

Note: This bit is reset by a system reset.

Bits 7:4 Reserved, must be kept at reset value.

Bit 3 **BRR**: Backup regulator ready

Set by hardware to indicate that the Backup Regulator is ready.

0: Backup Regulator not ready

1: Backup Regulator ready

Note: This bit is not reset when the device wakes up from Standby mode or by a system reset or power reset. This bit is reset by a backup domain reset.

Bit 2 **PVDO**: PVD output

This bit is set and cleared by hardware. It is valid only if PVD is enabled by the PVDE bit.

0: V_{DD} is higher than the PVD threshold selected with the PLS[2:0] bits.

1: V_{DD} is lower than the PVD threshold selected with the PLS[2:0] bits.

Note: The PVD is stopped by Standby mode. For this reason, this bit is equal to 0 after Standby or reset until the PVDE bit is set.

Bit 1 **SBF**: Standby flag

This bit is set by hardware and cleared only by a POR/PDR (power-on reset/power-down reset) or by setting the CSBF bit in the PWR_CR register.

0: Device has not been in Standby mode

1: Device has been in Standby mode

Bit 0 **WUF**: Wake-up flag

This bit is set by hardware and cleared either by a system reset or by setting the CWUF bit in the PWR_CR register.

0: No wake-up event occurred

1: A wake-up event was received from the WKUP pin or from the RTC alarm (Alarm A or Alarm B), RTC Tamper event, RTC TimeStamp event or RTC Wake-up).

Note: An additional wake-up event is detected if the WKUP pin is enabled (by setting the EWUP bit) when the WKUP pin level is already high.

5.5 Power control registers (STM32F42xxx and STM32F43xxx)

5.5.1 PWR power control register (PWR_CR) for STM32F42xxx and STM32F43xxx

Address offset: 0x00

Reset value: 0x0000 C000 (reset by wake-up from Standby mode)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved												UDEN[1:0]		ODSWEN	ODEN
												rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
VOS[1:0]		ADCD1	Res.	MRUDS	LPUDS	FPDS	DBP	PLS[2:0]			PVDE	CSBF	CWUF	PDDS	LPDS
rw	rw	rw		rw	rw	rw	rw	rw	rw	rw	rw	rc_w1	rc_w1	rw	rw

Bits 31:20 Reserved, must be kept at reset value.

Bits 19:18 **UDEN[1:0]**: Under-drive enable in stop mode

These bits are set by software. They allow to achieve a lower power consumption in Stop mode but with a longer wake-up time.

When set, the digital area has less leakage consumption when the device enters Stop mode.

00: Under-drive disable

01: Reserved

10: Reserved

11: Under-drive enable

Bit 17 **ODSWEN**: Over-drive switching enabled.

This bit is set by software. It is cleared automatically by hardware after exiting from Stop mode or when the ODEN bit is reset. When set, It is used to switch to Over-drive mode.

To set or reset the ODSWEN bit, the HSI or HSE must be selected as system clock.

The ODSWEN bit must only be set when the ODRDY flag is set to switch to Over-drive mode.

0: Over-drive switching disabled

1: Over-drive switching enabled

Note: On any over-drive switch (enabled or disabled), the system clock is stalled during the internal voltage set up.

Bit 16 **ODEN**: Over-drive enable

This bit is set by software. It is cleared automatically by hardware after exiting from Stop mode. It is used to enable the Over-drive mode in order to reach a higher frequency.

To set or reset the ODEN bit, the HSI or HSE must be selected as system clock. When the ODEN bit is set, the application must first wait for the Over-drive ready flag (ODRDY) to be set before setting the ODSWEN bit.

0: Over-drive disabled

1: Over-drive enabled

Bits 15:14 **VOS[1:0]**: Regulator voltage scaling output selection

These bits control the main internal voltage regulator output voltage to achieve a trade-off between performance and power consumption when the device does not operate at the maximum frequency (refer to the STM32F42xx and STM32F43xx datasheets for more details).

These bits can be modified only when the PLL is OFF. The new value programmed is active only when the PLL is ON. When the PLL is OFF, the voltage scale 3 is automatically selected.

00: Reserved (Scale 3 mode selected)

01: Scale 3 mode

10: Scale 2 mode

11: Scale 1 mode (reset value)

Bit 13 **ADCD1**:

0: No effect.

1: Refer to AN4073 for details on how to use this bit.

Note: This bit can only be set when operating at supply voltage range 2.7 to 3.6V and when the Prefetch is OFF.

Bit 12 Reserved, must be kept at reset value.

Bit 11 **MRUDS**: Main regulator in deepsleep under-drive mode

This bit is set and cleared by software.

0: Main regulator ON when the device is in Stop mode

1: Main Regulator in under-drive mode and Flash memory in power-down when the device is in Stop under-drive mode.

Bit 10 **LPUDS**: Low-power regulator in deepsleep under-drive mode

This bit is set and cleared by software.

0: Low-power regulator ON if LPDS bit is set when the device is in Stop mode

1: Low-power regulator in under-drive mode if LPDS bit is set and Flash memory in power-down when the device is in Stop under-drive mode.

Bit 9 **FPDS**: Flash power-down in Stop mode

When set, the Flash memory enters power-down mode when the device enters Stop mode.

This allows to achieve a lower consumption in stop mode but a longer restart time.

0: Flash memory not in power-down when the device is in Stop mode

1: Flash memory in power-down when the device is in Stop mode

Bit 8 **DBP**: Disable backup domain write protection

In reset state, the RCC_BDCR register, the RTC registers (including the backup registers), and the BRE bit of the PWR_CSR register, are protected against parasitic write access. This bit must be set to enable write access to these registers.

0: Access to RTC and RTC Backup registers and backup SRAM disabled

1: Access to RTC and RTC Backup registers and backup SRAM enabled

Bits 7:5 **PLS[2:0]**: PVD level selection

These bits are written by software to select the voltage threshold detected by the programmable voltage detector

000: 2.0 V
001: 2.1 V
010: 2.3 V
011: 2.5 V
100: 2.6 V
101: 2.7 V
110: 2.8 V
111: 2.9 V

Note: Refer to the electrical characteristics of the datasheet for more details.

Bit 4 **PVDE**: Programmable voltage detector enable

This bit is set and cleared by software.

0: PVD disabled
1: PVD enabled

Bit 3 **CSBF**: Clear standby flag

This bit is always read as 0.

0: No effect
1: Clear the SBF Standby Flag (write).

Bit 2 **CWUF**: Clear wake-up flag

This bit is always read as 0.

0: No effect
1: Clear the WUF Wake-up Flag **after 2 System clock cycles**

Bit 1 **PDDS**: Power-down deepsleep

This bit is set and cleared by software. It works together with the LPDS bit.

0: Enter Stop mode when the CPU enters deepsleep. The regulator status depends on the LPDS bit.
1: Enter Standby mode when the CPU enters deepsleep.

Bit 0 **LPDS**: Low-power deepsleep

This bit is set and cleared by software. It works together with the PDDS bit.

0: Main voltage regulator ON during Stop mode
1: Low-power voltage regulator ON during Stop mode

5.5.2 PWR power control/status register (PWR_CSR) for STM32F42xxx and STM32F43xxx

Address offset: 0x04

Reset value: 0x0000 0000 (not reset by wake-up from Standby mode)

Additional APB cycles are needed to read this register versus a standard APB read.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved												UDRDY[1:0]		ODSWRDY	ODRDY
												rc_w1	rc_w1	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res	VOSRDY	Reserved				BRE	EWUP	Reserved.				BRR	PVDO	SBF	WUF
	r					rw	rw					r	r	r	r

Bits 31:20 Reserved, must be kept at reset value.

Bits 19:18 **UDRDY[1:0]**: Under-drive ready flag

These bits are set by hardware when MCU entered stop Under-drive mode and exited. When the under-drive mode is enabled, these bits are not set as long as the MCU has not entered stop mode yet. They are cleared by programming them to 1.

00: Under-drive is disabled

01: Reserved

10: Reserved

11: Under-drive mode is activated in Stop mode.

Bit 17 **ODSWRDY**: Over-drive mode switching ready

0: Over-drive mode is not active.

1: Over-drive mode is active on digital area on 1.2 V domain

Bit 16 **ODRDY**: Over-drive mode ready

0: Over-drive mode not ready.

1: Over-drive mode ready

Bit 14 **VOSRDY**: Regulator voltage scaling output selection ready bit

0: Not ready

1: Ready

Bits 13:10 Reserved, must be kept at reset value.

Bit 9 **BRE**: Backup regulator enable

When set, the Backup regulator (used to maintain backup SRAM content in Standby and V_{BAT} modes) is enabled. If BRE is reset, the backup regulator is switched off. The backup SRAM can still be used but its content is lost in the Standby and V_{BAT} modes. Once set, the application must wait that the Backup Regulator Ready flag (BRR) is set to indicate that the data written into the RAM is maintained in the Standby and V_{BAT} modes.

0: Backup regulator disabled

1: Backup regulator enabled

Note: This bit is not reset when the device wakes up from Standby mode, by a system reset, or by a power reset. This bit is reset by a backup domain reset.

The DBP bit of the PWR_CR register must be set before BRE can be written.

Bit 8 **EWUP**: Enable WKUP pin

This bit is set and cleared by software.

0: WKUP pin is used for general purpose I/O. An event on the WKUP pin does not wake up the device from Standby mode.

1: WKUP pin is used for wake-up from Standby mode and forced in input pull down configuration (rising edge on WKUP pin wakes-up the system from Standby mode).

Note: This bit is reset by a system reset.

Bits 7:4 Reserved, must be kept at reset value.

Bit 3 **BRR**: Backup regulator ready

Set by hardware to indicate that the Backup Regulator is ready.

0: Backup Regulator not ready

1: Backup Regulator ready

Note: This bit is not reset when the device wakes up from Standby mode or by a system reset or power reset. This bit is reset by a backup domain reset.

Bit 2 **PVDO**: PVD output

This bit is set and cleared by hardware. It is valid only if PVD is enabled by the PVDE bit.

0: V_{DD} is higher than the PVD threshold selected with the PLS[2:0] bits.

1: V_{DD} is lower than the PVD threshold selected with the PLS[2:0] bits.

Note: The PVD is stopped by Standby mode. For this reason, this bit is equal to 0 after Standby or reset until the PVDE bit is set.

Bit 1 **SBF**: Standby flag

This bit is set by hardware and cleared only by a POR/PDR (power-on reset/power-down reset) or by setting the CSBF bit in the [PWR power control register \(PWR_CR\)](#) for [STM32F405xx/07xx and STM32F415xx/17xx](#)

0: Device has not been in Standby mode

1: Device has been in Standby mode

Bit 0 **WUF**: Wake-up flag

This bit is set by hardware and cleared either by a system reset or by setting the CWUF bit in the PWR_CR register.

0: No wake-up event occurred

1: A wake-up event was received from the WKUP pin or from the RTC alarm (Alarm A or Alarm B), RTC Tamper event, RTC TimeStamp event or RTC Wake-up).

Note: An additional wake-up event is detected if the WKUP pin is enabled (by setting the EWUP bit) when the WKUP pin level is already high.

5.6 PWR register map

The following table summarizes the PWR registers.

Table 32. PWR - register map and reset values for STM32F405xx/07xx and STM32F415xx/17xx

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x000	PWR_CR Reset value	Reserved																VOS	Reserved				FPDS	DBP	PLS[2:0]			PVDE	CSBF	CWUF	PDDS	LPDS	
0x004	PWR_CSR Reset value	Reserved																VOSRDY	Reserved				BRE	EWUP	Reserved			BRR	PVDO	SBF	WUF		
																		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

Table 33. PWR - register map and reset values for STM32F42xxx and STM32F43xxx

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0			
0x000	PWR_CR	Reserved													UDEN[1:0]		ODSWEN	ODEN	VOS[1:0]		ADGDC1	Reserved	MRUDS	LPUDS	FPDS	DBP	PLS[2:0]			PVDE	CSBF	CWUF	PDDS	LPDS		
	Reset value														0	0	0	0	1	1	0		0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x004	PWR_CSR	Reserved													UDRDY[1:0]		ODSWRDY	ODRDY	Reserved	VOSRDY	Reserved				BRE	EWUP	Reserved			BRR	PVDO	SBF	WUF			
	Reset value														0	0	0	0		0					0	0							0	0	0	0

Refer to [Section 2.3: Memory map](#) for the register boundary addresses.